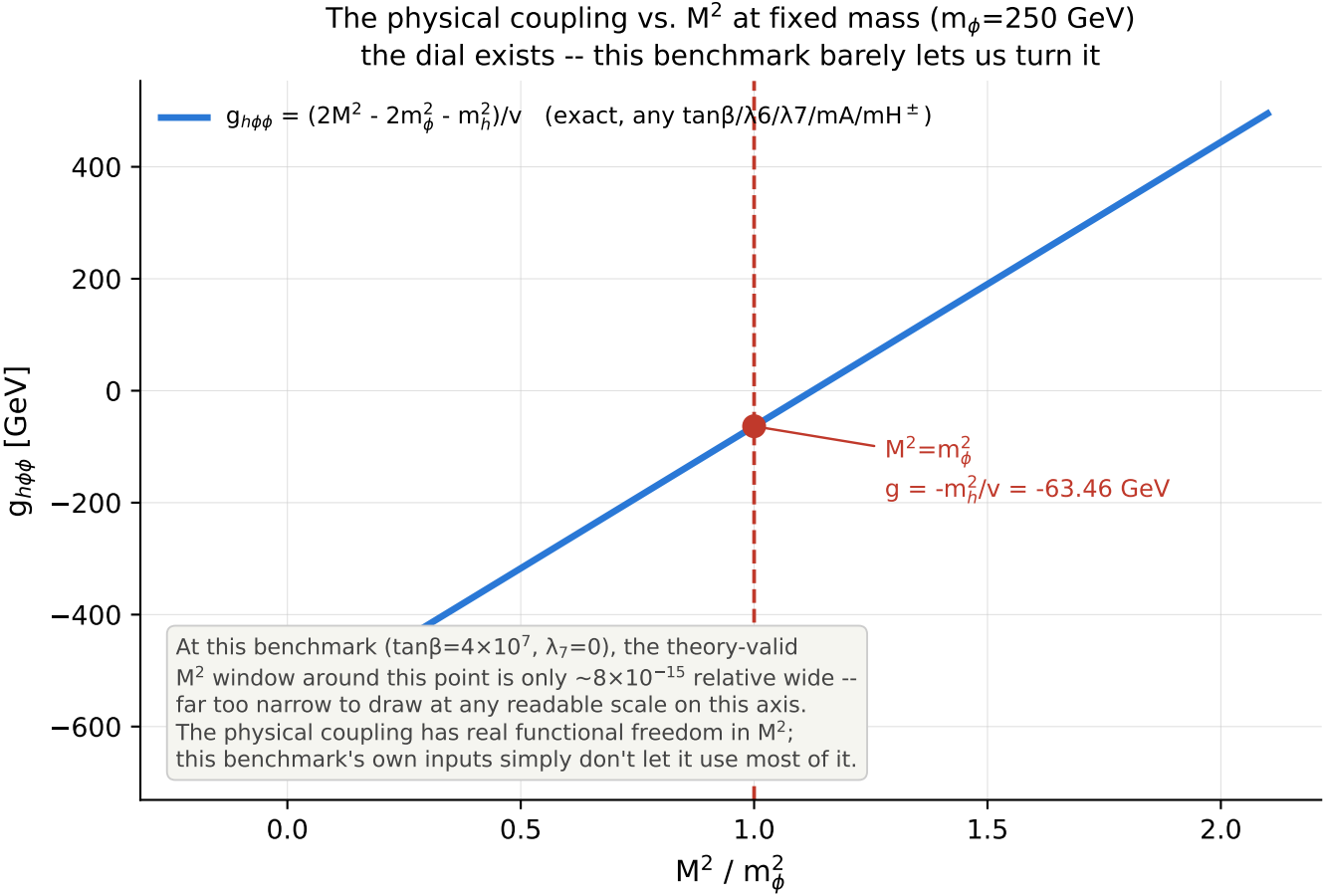
The physical coupling vs. M² at fixed mass (m_φ=250 GeV)
the dial exists -- this benchmark barely lets us turn it
g_hφφ = (2M² - 2m_φ² - m_h²)/v (exact, any tanβ/λ6/λ7/mA/mH±)
400
200
0
−200
−400
−600
g_hφφ [GeV]
M²=m_φ²
g = -m_h²/v = -63.46 GeV
At this benchmark (tanβ=4×10⁷, λ₇=0), the theory-valid
M² window around this point is only ~8×10⁻¹⁵ relative wide --
far too narrow to draw at any readable scale on this axis.
The physical coupling has real functional freedom in M²;
this benchmark's own inputs simply don't let it use most of it.
0.0
0.5
1.0
1.5
2.0
M² / m_φ²